



Targeted left ventricular lead placement in biventricular pacing for heart failure: a national, multicentre, double-blind, randomised controlled trial in Denmark

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Summary

Background Left ventricular lead position is a key determinant of optimal biventricular pacing for heart failure. Observational data indicate better outcomes in patients with late myocardial activation at the left ventricular lead site. We aimed to test whether targeting the left ventricular lead at site of latest electrical activation improves outcomes.

Methods In this national, randomised, controlled, double-blinded, superiority trial we included patients from all five Danish university centres performing device implantation who had heart failure and wide QRS on guideline-directed medication, eligible for biventricular pacing. We randomly assigned (1:1) patients to either the intervention group, with left ventricular lead placement at the latest electrically activated site within the coronary sinus branches, or to control group, with left ventricular lead placement in a posterolateral non-apical branch. Primary outcome was a composite of time to death or first unplanned heart failure hospitalisation. We used block randomisation and masked patients and all study personnel (except operating room personnel) to treatment allocation. The primary outcome was a composite of time to death or first unplanned hospitalisation for heart failure. Analysis followed a modified intention-to-treat approach. Safety was assessed in all correctly assigned patients. The trial is registered at ClinicalTrials.gov (NCT03280862) and is complete.

Findings From March 20, 2018, to June 3, 2024, we included 1001 patients (255 females and 746 males); 499 were randomly assigned to the intervention group and 502 to the control group. One incorrectly enrolled patient was excluded, leaving 1000 patients for the modified intention-to-treat analysis. Electrical activation at left ventricular lead was a mean of 9 milliseconds (95% CI 5–13) later in the intervention group. Follow-up was completed on Feb 27, 2026, after a median 45·8 months (IQR 28·5–65·7). The primary endpoint occurred in 139 (28%) of 499 patients in the intervention group and 128 (26%) of 501 in control group (hazard ratio 1·10 [95% CI 0·86–1·39]; $p=0·45$). Overall complication risk was similar between groups (71 [14%] patients in the intervention group vs 64 [13%] in the control group), but lead-related complications were more frequent in the intervention group. One procedure-related death occurred in the intervention group.

Interpretation In patients with heart failure and prolonged QRS, targeted left ventricular lead placement at the site of latest electrical activation did not reduce the composite endpoint of death or unplanned hospitalisation for heart failure compared with conventional lead placement in a posterolateral, non-apical position.

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Introduction

Biventricular pacing for cardiac resynchronisation therapy (CRT) in patients with heart failure with reduced ejection fraction and wide QRS effectively prolongs survival and reduces heart failure symptoms and hospitalisations,^{1–4} but it is well recognised that a considerable proportion of these patients have no measurable benefit from this treatment.⁵ This subgroup of patients, often called non-responders, cannot be

identified at time of implantation of the biventricular pacing system. Therefore, refinement of the implantation technique for biventricular pacing to improve the effect of CRT and have fewer patients with no benefit from this treatment is highly relevant.

The anatomical placement of the left ventricular lead is considered a main determinant of the positive effect of biventricular pacing. Typically, the left ventricular lead is placed in a posterolateral coronary sinus branch in a

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See Online for appendix 1

See Online for appendix 2

Research in context

Evidence before this study

No formal literature review was done before planning this study. Biventricular pacing for cardiac resynchronisation therapy (CRT) in patients with heart failure and broad QRS improves left ventricular ejection fraction, functional status, and quality of life; reduces heart failure hospitalisations; and prolongs longevity. However, a considerable portion of these patients have no measurable benefit from CRT. A series of observational reports indicated that the more delayed the local electrical activation was at the site of the left ventricular lead, the better the outcomes of the patients with heart failure and bundle-branch block treated with CRT. Small randomised controlled trials (RCTs) investigating targeting the left ventricular lead towards the latest activation in the left ventricle showed various results, but differences in methods and endpoints used precluded a strong meta-analysis. We did two single centre RCTs testing targeted left ventricular lead placement in CRT for heart failure, including 182 and 122 patients, respectively. Results indicated that after 6 months of follow-up, targeted placement of the left ventricular lead towards the latest activated segment of the left ventricle was associated with a higher rate of response to CRT and potentially a higher increase in left ventricular ejection fraction. We therefore joined national centres for this double-blind, long-term follow-up RCT, which was powered to show whether

targeted placement of the left ventricular lead in CRT for heart failure reduces the clinically relevant endpoint of death or unplanned hospitalisation for heart failure and thus might reduce the proportion of patients not responding to CRT.

Added value of this study

This large, double-blind RCT shows that in patients with heart failure and prolonged QRS, targeted placement of the left ventricular pacing lead at the site of latest electrical activation has no effect in reducing the risk of death or unplanned hospitalisation for heart failure compared with conventional placement of the lead in a posterolateral, non-apical position. Numerically, more lead-related complications were observed with the intervention. This was the largest trial to date on this topic.

Implications of all the available evidence

The findings in this trial, when added to current evidence, indicate that for routine implantation of CRT for heart failure, the left ventricular lead should be placed in a posterolateral, non-apical position rather than guided by electrical mapping. Placing the lead guided by electrical mapping makes no difference to outcomes, but requires longer operation time, requires longer fluoroscopy time, and results in more lead-related complications.

non-apical position. However, observational data indicate that patients with left ventricular leads placed in regions of later activation derive greater benefit from biventricular pacing, with larger improvements in left ventricular systolic function and improved survival.^{6–9} Small randomised controlled trials investigating whether targeting the left ventricular lead at late-activated segments of the left ventricle have shown varying results.^{10–18} Electrical activation time of the left ventricular segments adjacent to coronary sinus branches, where a left ventricular pacing lead can be positioned, is easily measured during the implantation procedure.^{19,20}

We aimed to test whether targeting the left ventricular lead at the site of latest electrical activation in the coronary sinus branches improves measured outcomes, such as time to death or unplanned hospitalisation for heart failure in patients with an indication for cardiac resynchronisation therapy.

Methods

Study design

The DANISH-CRT trial was an investigator-initiated, double-blinded, randomised, controlled, superiority trial done in all five Danish university centres performing CRT device implantations. The trial design and rationale were published previously,²¹ and the full protocol, including statistical analysis plan and protocol amendments made during the study, are presented in

appendix 2. The trial was designed and overseen by a steering committee. The trial protocol was approved by the Scientific Ethics Committee of Region Midtjylland of Denmark (1–10–72–330–16) and the Danish Data Protection Agency (1–16–02–660–16). We followed the principles of the Declaration of Helsinki. Independent Good Clinical Practice monitoring was provided for a random subset of patients by the Good Clinical Practice unit of Aarhus University. No patients were involved in the design, conduct, or reporting of the trial. The trial is registered on ClinicalTrials.gov (NCT03280862) and is complete.

Patients

We included patients with symptomatic heart failure in New York Heart Association (NYHA) functional class II, III, or outpatient IV; a left ventricular ejection fraction (LVEF) of 35% or less and a wide QRS complex due to left bundle-branch block (LBBB) equal to or wider than 130 milliseconds; other bundle-branch block (BBB) wider than 150 milliseconds; right ventricular pacing over 50% of the time; or an indication for right ventricular pacing with a high expected percentage of right ventricular pacing.²² Patients were referred for primary implantation of CRT with or without a concomitant implantable cardioverter defibrillator backup, or for a device upgrade from right ventricular pacing. Patients were included in the trial after referral for CRT and before the implantation

procedure. Only patients with a life expectancy longer than 2 years were eligible for inclusion. Patients were required, whenever possible, to be treated with guideline-specified target doses of an angiotensin converting-enzyme inhibitor, an angiotensin 2 receptor blocker, or a combined angiotensin 2 receptor blocker and neprilysin inhibitor; a β blocker; and an aldosterone receptor antagonist. After release of the 2021 European Society of Cardiology Guidelines for treatment of heart failure,²³ a sodium-glucose cotransporter 2 inhibitor was initiated at inclusion in the trial, or at first follow-up in patients already included. Patients with a myocardial infarction or coronary bypass artery grafting surgery within 3 months or participating in another experimental trial were excluded. Detailed description of inclusion and exclusion criteria are provided in the protocol in appendix 2 (pp 4–5). Data on sex (female or male) were obtained by self-report. Data on ethnicity were not collected. All patients gave written informed consent.

Randomisation and masking

Patients were randomly allocated (1:1) to the intervention group or control group. Information about treatment allocation was available in the case-record form exclusively for the implanting physician after the patient had been included. Randomisation was web-based and integrated in the case-record form. Randomisation was stratified for baseline QRS morphology (LBBB, other BBB, right ventricular pacing, or indication for right ventricular pacing). We used block-randomisation (blocks of ten plus ten patients for LBBB and right ventricular pacing, five plus five patients for other groups) to ensure balanced allocation of patients to each treatment group within each centre. The trial was double blinded as only the implanting physician could access information in the case-record form about treatment allocation to provide the assigned treatment. Treatment allocation was also disclosed to operating room personnel. This information was not accessible to other personnel involved in the patient's treatment, care, or study follow-up. Neither the implanting physician nor the operating room personnel were involved in the follow-up of the patients. Treatment allocation was not disclosed to patients before or during device implantation or during follow-up. To assess whether masking could be considered successful, we did a survey of 148 patients attending 6-month or 24-month follow-up and the responsible personnel in one of the five participating centres during 2022–23. Patients and responsible personnel were asked whether they knew which of the two treatment groups they were assigned to.

Procedures

In the intervention group, local electrical activation was measured and recorded in all coronary sinus branches where a left ventricular lead could potentially be placed using the left ventricular lead and/or an electrically active

guidewire. Mapping was done during intrinsic ventricular conduction in patients with LBBB, and during right ventricular pacing in the other three strata of patients (other BBB, right ventricular pacing, or indication for right ventricular pacing). The left ventricular lead was positioned in the coronary sinus branch according to the site of latest local electrical activation if lead stability, pacing threshold, and threshold for phrenic nerve stimulation were acceptable. In the control group, electrical mapping was not done, and the left ventricular lead was positioned according to contemporary practice, preferentially in a posterolateral, non-apical position. The final left ventricular activation delay was measured by the implanting physician after the implantation procedure. In all patients, right ventricular and atrial pacing leads were placed in the right-sided chambers according to local standard care. The atrioventricular interval was programmed to ensure biventricular pacing during sinus rhythm.

Timing between left and right ventricular pacing was optimised via steps of 20 milliseconds in interventricular pacing intervals to yield the shortest QRS complex, and post-implantation programming of pacing and defibrillation parameters according to local clinical practice. Use of algorithms for automatic optimisation of the atrioventricular delay to promote fusion of intrinsic conduction and left ventricular pacing were not used in the trial, nor was left ventricular pacing only allowed. Cardiac CT was done no later than 3 months after implantation to determine lead positions.

Follow-up was identical in the two treatment groups, consisting of follow-up visits after 3, 6, and 12 months, and annually thereafter until the end of the trial. Detailed information on assessments done at each follow-up visit are available in appendix 2 (pp 10–11).

Outcomes

The primary outcome was the composite of time to death or first unplanned hospitalisation for heart failure. Unplanned hospitalisation for heart failure was defined as acute hospitalisation lasting more than 24 h (or an overnight stay) of a patient who presents with new or worsening symptoms of heart failure, objective evidence of new or worsening heart failure, and initiation or intensification of heart failure-specific treatment. This treatment included intensified diuretic treatment, initiation of a new guideline-directed drug for heart failure, or dosage-increase in ongoing guideline-directed heart failure medication, as well as inotropes and intensive-care treatment for heart failure.

Secondary outcomes were the two components of the primary outcome, assessed individually, and time to cardiovascular death, sudden death, and atrial fibrillation, defined as clinically diagnosed atrial fibrillation or device-detected atrial fibrillation longer than 30 s documented during follow-up. In addition, we assessed changes from baseline in left ventricular end-diastolic

	Intervention group (n=499)	Control group (n=501)
Age, years	73 (65–78)	72 (63–77)
Sex		
Female	130 (26%)	125 (25%)
Male	369 (74%)	376 (75%)
Height*, cm	176 (169–182)	175 (169–181)
Weight*, kg	83 (74–96)	84 (74–97)
QRS duration, milliseconds	164 (158–180)	166 (158–180)
Left ventricular ejection fraction, %	27% (6)	27% (6)
Left ventricular end-diastolic volume, mL	187 (73)	190 (64)
Left ventricular end-systolic volume, mL	138 (62)	140 (54)
Estimated GFR, mL/min per 1.73m ²		
0–29	20 (4%)	21 (4%)
30–60	206 (41%)	196 (39%)
>60	269 (54%)	279 (56%)
NYHA class		
II	279 (56%)	286 (57%)
III	216 (43%)	206 (41%)
IV	2 (<1%)	9 (2%)
Heart failure causes†		
Ischaemic heart disease	221 (44%)	193 (39%)
Non-ischaemic dilated cardiomyopathy	271 (54%)	284 (57%)
Valvular heart disease	37 (7%)	44 (9%)
Hypertrophic cardiomyopathy	21 (4%)	26 (5%)
Arrhythmogenic cardiomyopathy	16 (3%)	9 (2%)
Congenital heart disease	4 (<1%)	1 (<1%)
Cardiac sarcoidosis	7 (1%)	5 (1%)
Pacing-induced heart failure	87 (17%)	87 (17%)
Pre-existing cardiac pacemaker or implantable cardioverter defibrillator	174 (35%)	174 (35%)
Indication for biventricular pacing		
Left bundle-branch block with QRS ≥130 milliseconds	336 (67%)	342 (68%)
Other bundle-branch block with QRS >150 milliseconds	43 (9%)	41 (8%)
Paced QRS and need for ventricular pacing	111 (22%)	111 (22%)
Indication for pacing and low ejection fraction	9 (2%)	7 (1%)

(Table 1 continues in next column)

and end-systolic volumes and LVEF measured by echocardiography after 6 months, changes in NYHA class and quality of life measured as the Clinical Summary Score of the Kansas City Cardiomyopathy Questionnaire (KCCQ-12) from baseline to 6-month follow-up, and 6-min walk test after 6 months. Safety endpoints included procedure-related and device-related complications and implantable cardioverter defibrillator therapy (shock or anti-tachycardia pacing).

All echocardiograms were analysed in a central core laboratory by people that were masked to treatment

	Intervention group (n=499)	Control group (n=501)
(Continued from previous column)		
Coexisting conditions		
Hypertension	294 (59%)	289 (58%)
Persistent or permanent atrial fibrillation	96 (19%)	79 (16%)
Paroxysmal atrial fibrillation	108 (22%)	119 (24%)
Chronic obstructive pulmonary disease	55 (11%)	62 (12%)
Diabetes	120 (24%)	135 (27%)
Medication		
ACE inhibitor, angiotensin 2 receptor blocker, or ARNI	465 (93%)	475 (95%)
β blocker	434 (87%)	429 (86%)
Mineralocorticoid receptor blocker	349 (70%)	337 (67%)
SGLT2 inhibitor	194 (39%)	191 (38%)
Diuretics	345 (69%)	326 (65%)
Digoxin	21 (4%)	20 (4%)
Amiodarone	58 (12%)	51 (10%)
Nt-Pro BNP†, ng/L	1259 (574–5–2672)	1260 (487–2666)

Data are median (IQR), mean (SD), or n (%). ACE=angiotensin-converting enzyme. ARNI=angiotensin receptor-neprilysin inhibitor. GFR=glomerular filtration rate. Nt-Pro BNP=N-terminal pro B-type Natriuretic Peptide. NYHA=New York Heart Association. SGLT2=sodium-glucose co-transporter 2. *Missing values for baseline characteristics were less than 2% of patients except for the variables of height (n=37; 3.7%), weight (n=50; 5.0%), and Nt-Pro BNP (n=244; 24.4%). †Some patients might have more than one cause for heart failure.

Table 1: Baseline characteristics

allocation. An endpoint committee comprising three experienced cardiologists and trialists not otherwise involved in the trial adjudicated all hospitalisations during follow-up to determine whether the prespecified criteria were met for the endpoint of unplanned heart failure hospitalisation, and assessed all deaths to classify cardiac deaths and sudden deaths according to prespecified criteria. Criteria for these endpoints are detailed in appendix 2 (pp 4–5). The endpoint committee was masked to group allocation.

Statistical analysis

Based upon an 182-patient randomised, double-blinded, pilot study showing a highly significant reduction in non-responders to CRT with targeted left ventricular lead placement^{9,10} combined with findings in two other early studies on targeted left ventricular lead implantation,^{12–14} we assumed a relative 25% reduction in the primary endpoint from absolute 35% to 26% to be realistic with our intervention. The study was designed to have 80% power to detect a 25% difference in the primary composite endpoint between treatment groups. At least 264 primary outcome events were required for the study to be conclusive. The sample size calculation was done for a dichotomous endpoint, and the 0.75 relative risk expected in the intervention group is equivalent to a

detectable hazard ratio close to 0·7 when using time-to-event methods for group comparisons, which was the primary endpoint. With an expected attrition rate of less than 10%, we planned to include 1000 patients.²¹

All analyses were assessed in the modified intention-to-treat population, including all patients who were correctly enrolled and randomised. Baseline characteristics were presented as means with SDs or medians with IQRs, as appropriate. We analysed outcomes with time-to-event methods. We generated Kaplan–Meier curves for the primary outcome and all-cause deaths and comparisons were done with log-rank tests. We calculated cumulative incidence curves for unplanned hospitalisation for heart failure, cardiovascular death, sudden death, and atrial fibrillation to account for death or other causes of death as a competing risk, and we did comparisons via a competing risk regression analysis. We calculated hazard ratios with 95% CIs via unadjusted Cox regressions. Mean differences in echocardiographic parameters were compared with a two-sample Student's *t* test. We used risk ratios to estimate between-group differences in complication risk, assuming similar follow-up duration between groups.

For the primary outcome, a prespecified adjusted analysis was done with adjustment for age, sex, QRS morphology (LBBB, other BBB, right ventricular paced, or indication for right ventricular pacing), QRS width (≥ 150 milliseconds or < 150 milliseconds), NYHA class (2, 3, or outpatient 4), underlying disease (ischaemic heart disease or non-ischaemic heart disease), and centre. Proportional hazard assumptions were determined by Schoenfeld residuals, and minor violations were found for the adjusted model. This model was therefore stratified according to QRS width and centre to allow for different baseline hazard for these variables. Prespecified subgroup analyses of the primary outcome were completed for the baseline variables: age (≤ 72 years or > 72 years), sex, atrial fibrillation, upgrade from right ventricular pacing, underlying disease (ischaemic heart disease or non-ischaemic heart disease), local left ventricular activation time at left ventricular implantation site (≤ 134 milliseconds or > 134 milliseconds), and QRS morphology (LBBB, paced QRS, or other BBB). More than 20% of patients had missing data in the secondary outcome variables of six-min walk test distance and KCCQ-12 score. For six-min walk test distance, data were considered missing not at random, as the most common reasons for not completing the six-min walk test was reported to be “not able to walk” or “in too poor condition to complete”. Therefore, multiple imputation was not done for this variable. For KCCQ-12 score, multiple ($n=25$) imputation was done with chained equations and predictive mean matching in patients not deceased before their 6-month follow-up, including the baseline variables of age, sex, treatment group, baseline KCCQ-12 score, ischaemic heart disease, LVEF, pacing-induced heart failure, and NYHA class. A complete-case analysis

was also done. A two-sided $p < 0.05$ was considered statistically significant. No adjustment for multiple testing was applied. Analyses were done with R (version 4.4.2).

An independent Data Safety and Monitoring Board followed the trial during conduct and advised the steering committee. The committee charter is provided in appendix 2 (pp 13–14).

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

From March 20, 2018, to June 3, 2024, we included 1001 patients at five centres (255 females and 746 males). 499 patients were randomly allocated to the intervention

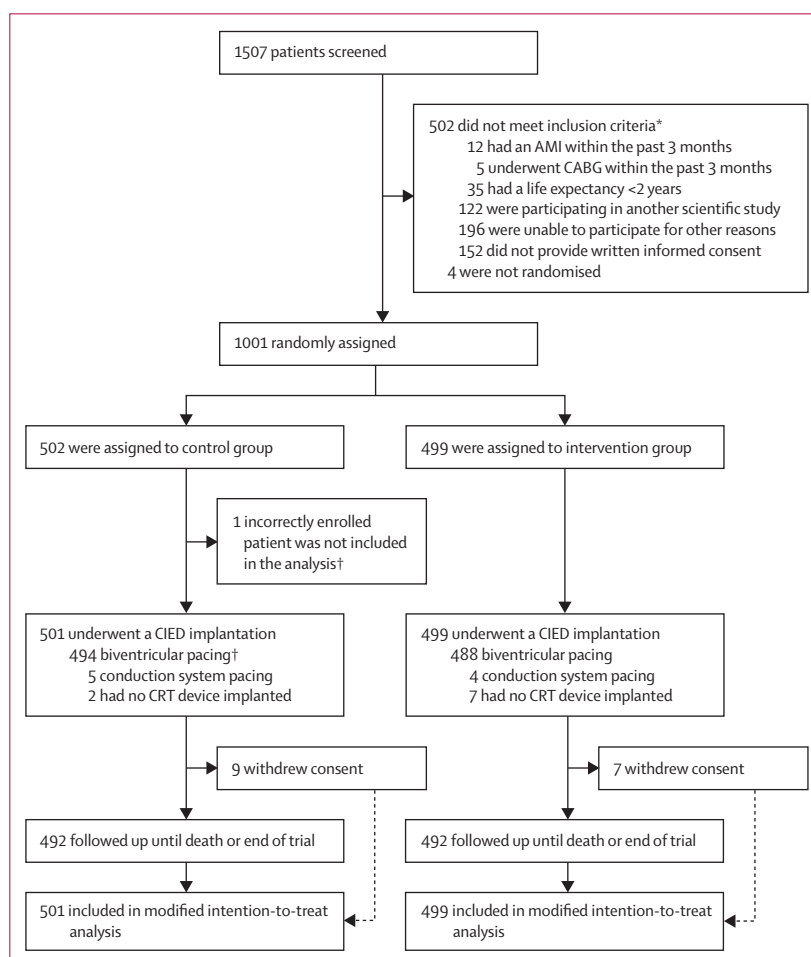


Figure 1: Patient inclusion flowchart

AMI=acute myocardial infarction. CABG=coronary artery bypass grafting. CIED=cardiac implantable electronic device. *For some patients, more than one criteria were not met. †Underwent randomisation despite not meeting the inclusion criteria (left ventricular ejection fraction $> 35\%$); no device was implanted and the participant was not followed up in the trial. ‡In one patient, implantation of a transvenous left ventricular lead was unsuccessful; instead, an epicardial left ventricular pacing lead was implanted thoracoscopically.

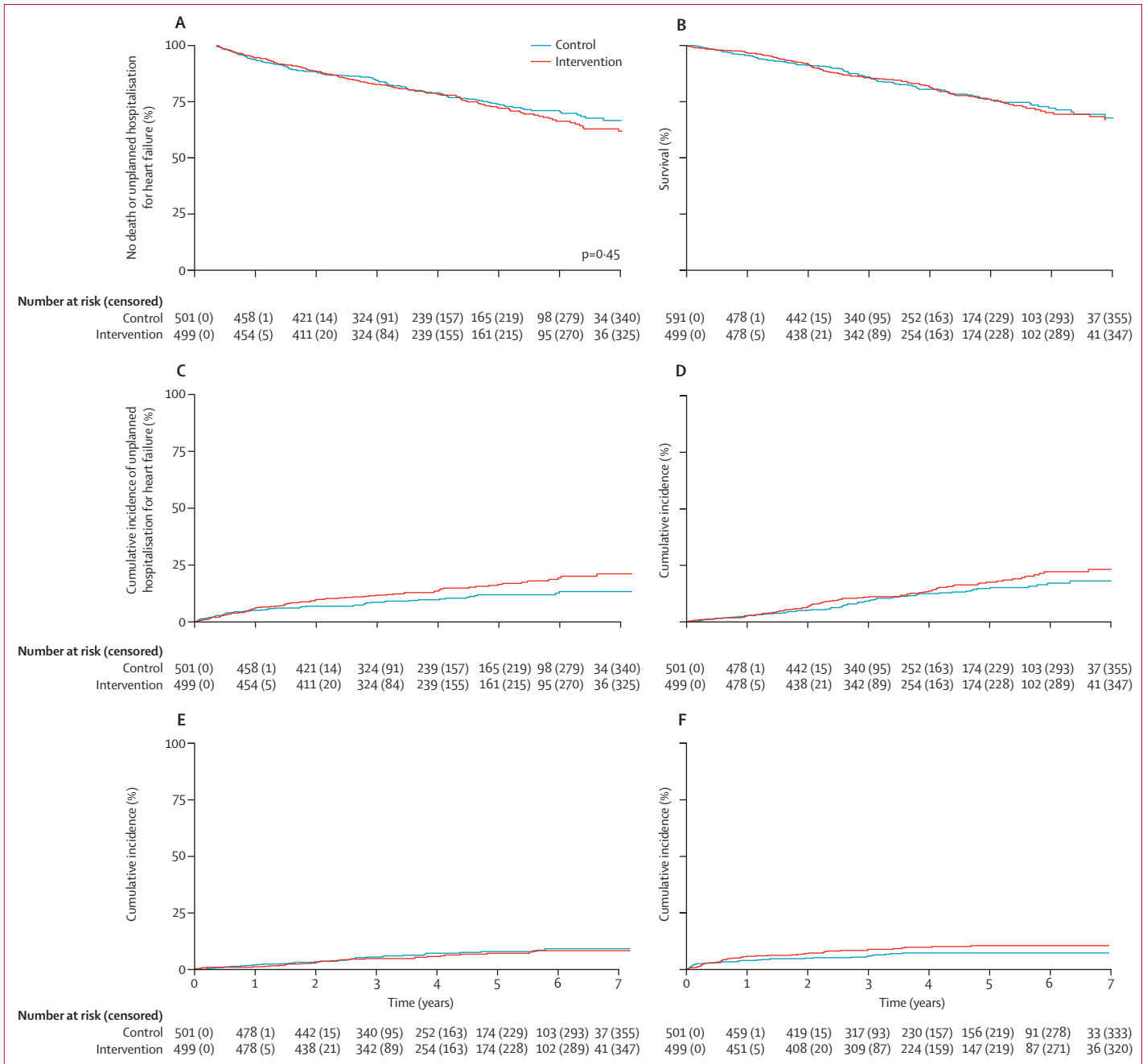


Figure 2: Time-to-event and curves for primary and secondary outcomes
Time-to-event curves as Kaplan-Meier curves for the primary composite outcome event of death or unplanned hospitalisation for heart failure (A) and for all-cause death (B). For the outcome unplanned hospitalisation for heart failure, there are cumulative incidence curves accounting for all-cause death as a competing risk (C). For the outcomes cardiovascular death (D), sudden death (E), and atrial fibrillation (Panel F), cumulative incidence curves accounting for death or other causes of death as competing risk are shown.

group and 502 patients to the control group. One patient was incorrectly enrolled and randomised despite preserved ejection fraction but underwent no device implantation or follow-up and was excluded from analysis, leaving 1000 patients for the modified intention-to-treat analysis. In 678 (67%) of 1000 patients, the indication for CRT was LBBB and wide QRS and

348 (35%) of 1000 patients underwent upgrade from right ventricular pacing. The two groups were overall well balanced on baseline characteristics (table 1). Baseline medications for heart failure were accordant with guideline recommendations in most patients. At baseline, 385 (39%) of 1000 patients were on sodium-glucose cotransporter 2 inhibitor treatment, and this was

initiated during follow-up in an additional 196 (20%) of 1000 patients. Follow-up was completed on Feb 27, 2026, resulting in a median follow-up period of 45·8 months (IQR 28·5–65·7). No patients were lost to follow-up for the primary outcome. No protocol derivations occurred during the trial.

In the survey of masking success, 13 (9%) of 148 patients replied that they knew which of the two treatment groups they were assigned to, and nine (6%) were able to correctly answer which group they were assigned, hereof six patients from the intervention group and three from the control group. None of the personnel doing follow-up were aware of treatment allocation.

Implantation of biventricular pacing was successful in 982 (98%) of 1000 patients (figure 1; appendix 2 pp 15–16). Of the 18 patients not treated with biventricular pacing, nine patients had conduction system pacing and nine had no CRT system implanted. In 486 (97%) of 499 patients in the intervention group, mapping of a median of three (IQR 2–4) coronary sinus branches was completed: in 414 (83%) patients the left ventricular lead was placed in the coronary sinus branch with the latest electrical activation, and in 389 (78·0%) of the 499 patients in the intervention group, the lead was placed at the site with the latest activation measured within the branch (appendix 2 pp 15–16). In the control group, 465 (93%) of 501 patients had the left ventricular lead placed according to the site judged by the operator to be optimum. In both treatment groups, most left ventricular leads were placed with the tip in a posterior or lateral segment, most commonly in a midventricular position. The anatomical placement of the left ventricular lead tip in the two groups is depicted graphically in appendix 2 (p 17). Activation time measured at left ventricular lead placement was a mean of 9 milliseconds (95% CI 5–13) longer in the intervention group (appendix 2 pp 15–16). QRS width at discharge was similar in the two groups, but the intervention group had longer procedure times (median 100 min [IQR 85–130] in the intervention group vs 81 min [63–120] in the control group) and fluoroscopy times (22 min [15–30] in the intervention group vs 15 min [9–25] in the control group), both $p < 0·0001$ between groups, and higher radiation dose (median dose-area product 19 Gy cm^2 [IQR 12–33] vs 14 Gy cm^2 [8–30]). Optimisation timing between left and right ventricular pacing in the treatment groups are shown in appendix 2 (p 18).

The primary composite outcome of death or first unplanned hospitalisation for heart failure occurred in 139 (28%) of 499 patients in the intervention group (7·1 events per 100 person-years) and 128 (26%) of 501 patients in the control group (6·5 events per 100 person-years). The hazard ratio for the primary endpoint in the intervention group as compared with the control group was 1·10 (95% CI 0·86–1·39; $p = 0·45$; figure 2; table 2). The prespecified adjusted analysis of the primary endpoint was done as a complete-case analysis including all 981 patients who had complete data

in all variables that were included in the adjusted analysis. This analysis thus included 136 (28%) events occurring in 491 (98%) patients in the intervention group and 121 (25%) events occurring in 490 (98%) patients in the control group and revealed similar results, with a hazard ratio of 1·06 (95% CI 0·82–1·36). The rate of unplanned hospitalisation for heart failure was higher in the intervention group, whereas no differences were observed between treatment groups for death, cardiovascular death, sudden death, or atrial fibrillation (figure 2; table 2). Ventricular remodelling as assessed by echocardiography after 6 months compared with baseline showed no difference between groups, as did the increase

	Intervention group (n=499)	Control group (n=501)	Effect measure (95% CI)	p value
Clinical outcomes*				
Death or unplanned hospitalisation for heart failure	139 (28%)	128 (26%)	1·10 (0·86 to 1·39)	0·45
Death	113 (23%)	110 (22%)	1·03 (0·79 to 1·34)	0·82
Cardiovascular death	82 (16%)	67 (13%)	1·23 (0·89 to 1·70)	0·21
Sudden cardiac death	32 (6%)	36 (7%)	0·90 (0·56 to 1·44)	0·65
Unplanned hospitalisation for heart failure	77 (15%)	53 (11%)	1·47 (1·03 to 2·08)	0·03
Atrial fibrillation	48 (10%)	34 (7%)	1·44 (0·93 to 2·23)	0·11
Six-month outcomes†‡				
Left ventricular end-diastolic volume, mL	154 (69)	154 (58)	−0·4 (−7·7 to 8·6)	0·92
Left ventricular end-systolic volume, mL	95 (57)	94 (49)	−0·8 (−7·6 to 5·9)	0·81
Left ventricular ejection fraction, %	41% (9·9)	41% (9·8)	0·2 (−1·1 to 1·4)	0·40
Increase in LVEF from baseline, %	14% (9·1)	14·2% (8·9)	−0·1 (−1·3 to 1·0)	0·81
Six-min walk test distance, m	436 (107)	432 (113)	4·2 (−11·5 to 19·9)	0·60
NYHA class	..	..	..	0·66
I	167 (33%)	154 (31%)	..	..
II	262 (53%)	282 (56%)	..	..
III	43 (9%)	41 (8%)	..	..
IV	1 (<1%)	2 (<1%)	..	..
Change in KCCQ-12 CSS score from baseline	11·9 (17·4)	11·0 (17·5)	0·9 (−1·8 to 3·6)	0·50
Difference in KCCQ-12 CSS with multiple imputation	..	..	0·6 (−1·8 to 3·1)	0·62
Safety outcomes§				
Patients with device-related complications				
At implantation	17 (3%)	14 (3%)	..	0·58
After implantation before discharge	21 (4%)	21 (4%)	..	0·99
During follow-up	39 (8%)	35 (7%)	..	0·62
Total patients with any complication	71 (14%)	64 (13%)	1·11 (0·81 to 1·52)	0·50
Patients with major complications				
Patients with one complication	48 (10%)	46 (9%)	..	..
Patients with two complications	10 (2%)	4 (1%)	..	..
Patients with three complications	1 (<1%)	0	..	..
Total patients with lead-related complications¶	34 (7%)	23 (5%)	1·48 (0·89 to 2·48)	0·13
Total patients with lead-related complications including perforations	39 (8%)	26 (5%)	1·51 (0·93 to 2·44)	0·09

(Table 2 continues on next page)

	Intervention group (N=499)	Control group (N=501)	Effect measure (95% CI)	p value
(Continued from previous page)				
Total patients with left ventricular lead-related complications	15 (3%)	6 (1%)	2.51 (0.98 to 6.42)	0.05
Patients with appropriate ICD therapy during follow-up**	36 (7%)	34 (7%)	1.06 (0.68 to 1.67)	0.79
Patients with inappropriate ICD-therapy during follow-up	8 (2%)	9 (2%)	0.89 (0.35 to 2.29)	0.81
Patients with inappropriate ICD-therapy caused by atrial tachyarrhythmia	5 (1%)	9 (2%)	..	..
Patients with inappropriate ICD-therapy caused by oversensing	3 (1%)	0	..	..

Data are n (%) or mean (SD), unless otherwise specified. KCCQ-12 CSS=Clinical Summary Score of the 12-item Kansas City Cardiomyopathy Score. LVEF=left ventricular ejection fraction. NYHA=New York Heart Association. ICD=implantable cardioverter defibrillator. *Effect measures for these outcomes are hazard ratios. †Effect measures for these outcomes are mean differences. ‡For the following variables, data were missing for some of the patients, because they did not survive to the 6-month follow-up or for other reasons: data were missing on LVEF for 56 (6%) patients, for left ventricular volumes for 57 (6%) patients, for six-min walk test distance for 242 (24%) patients, for NYHA-class for 48 (5%) patients, and for change in KCCQ-12 CSS for 362 (36%) patients. §Effect measures for these outcomes are risk ratios. ¶Includes lead dislodgements and high lead threshold or phrenic nerve stimulation not resolved by reprogramming. ||Includes lead dislodgements, tamponade, perforation, and high lead threshold or phrenic nerve stimulation not resolved by reprogramming. **Includes any ICD therapy, ICD shock, or anti-tachycardia pacing.

Table 2: Outcomes and adverse events

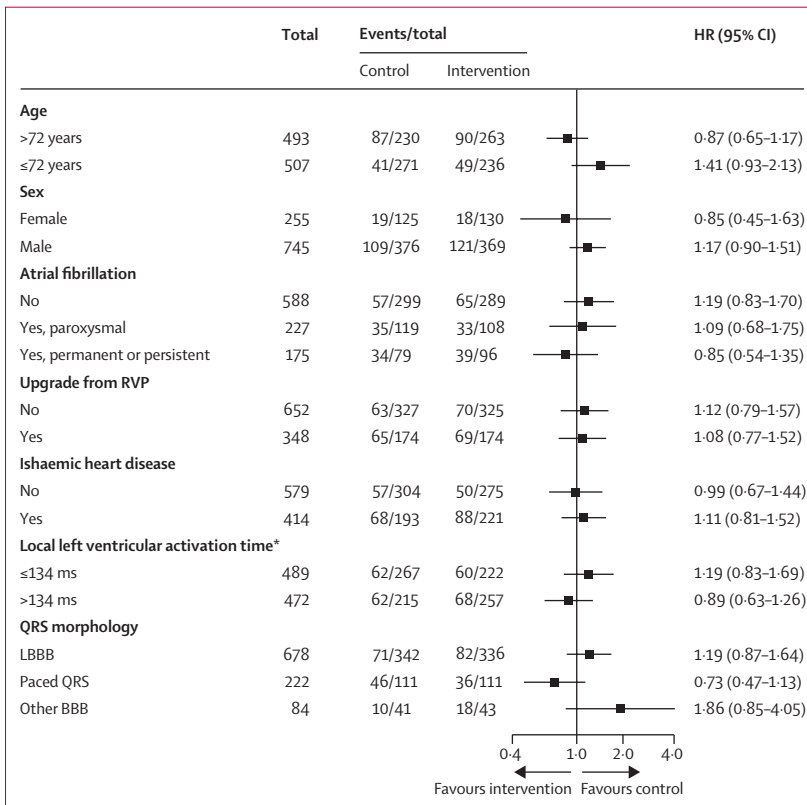


Figure 3: Forest plot of subgroup analyses

Predefined subgroup analyses of the primary composite outcome. BBB=bundle-branch block. LBBB=left bundle-branch block. RVP=right ventricular pacing. *Local left ventricular activation time measured from right ventricular activation.

in quality of life, NYHA class, and 6-min walk test done at 6 months (table 2). No differences in treatment effect were observed across any predefined subgroup analyses of the primary composite outcome (figure 3).

The incidence of device-related complications did not differ between groups, but more lead dislodgements were observed in the intervention group (table 2; table 3). One perioperative death occurred in the intervention group. Appropriate and inappropriate implantable cardioverter defibrillator therapy during follow-up occurred with similar proportions in the treatment groups (table 2).

Discussion

In patients with heart failure and prolonged QRS, positioning the left ventricular lead at the site of the latest electrical activation within the coronary sinus branches did not prolong time to the composite endpoint of death or unplanned hospitalisation for heart failure. Consistent findings were observed across secondary outcomes on ventricular remodelling and patient quality of life, functional status and physical capacity, and in the subgroup analyses of the primary endpoint. Numerically, more lead-related complications were observed in the intervention group, in addition to longer operation and fluoroscopy times. Our findings do not support routine electrical mapping to guide left ventricular lead placement in contemporary CRT practice.

Most patients have multiple coronary sinus branches suitable for left ventricular lead placement. The optimum strategy for branch selection in biventricular pacing has not previously been evaluated in a large, randomised trial. Observational data favour positioning the left ventricular lead in a posterolateral and non-apical location and targeting a site of late local activation.^{8,24,25} In this trial, the left ventricular lead was successfully positioned at the site of the latest electrical activation in four of every five patients at the cost of significantly longer procedure and fluoroscopy times, excess radiation dose, and a numerical excess of complications in the intervention group, driven primarily by more lead-related complications. Late-activated target sites in the intervention group were predominantly located in posterolateral segments of the left ventricle.

As intended, local activation at the targeted left ventricular pacing site occurred later in the intervention group than in the control group. Nevertheless, the additional delay at the final lead position was modest, with a mean of only 9 milliseconds, suggesting little added value of electrical mapping over conventionally selected optimal anatomical targets. Consistent with this interpretation, left ventricular lead locations were very similar between groups as assessed on cardiac CT, and no differences were observed in ventricular activation time after device optimisation, as reflected in QRS width. The observation of numerically more lead-related complications in the intervention group combined with

statistically more unplanned heart failure hospitalisations and marginally more patients with atrial fibrillation in the intervention group raises the question whether the intervention could even be harmful. The excess lead-related complications, although not statistically significant due to low absolute numbers, were likely related to more leads in the intervention group not being left in a wedged position to prioritise pacing from the latest activated site. In terms of the higher number of unplanned hospitalisations for heart failure and atrial fibrillation with the intervention, the trial was not powered for most secondary outcomes, and findings for secondary outcomes are exploratory in nature. Furthermore, we did not apply correction for multiple testing, which increases the risk of chance findings in secondary endpoints. We did not use active fixation left ventricular leads in the trial, and it is unknown whether use of such leads would result in less left ventricular lead dislocations.

In retrospect, the assumed effect size in the sample size estimation might have been overly optimistic. Nevertheless, this trial provides a clear answer to the predefined research question posed. At the cost of a non-negligible increase in operation time, fluoroscopy time, and radiation dose, and a clear numerical increase in lead-related complications in the intervention group, the changes in electrical delay with electrically targeted left ventricular lead placement were small and the pattern of left ventricular lead placement close to identical in the two treatment groups. Findings for the secondary endpoints, such as improvement after 6 months in LVEF, NYHA class, and quality-of-life score as well as 6-min walk test distance, were close to identical for the two treatment groups, supporting strongly the neutral result in the primary endpoint. It is therefore unlikely that targeted left ventricular lead placement would result in a large effect on mortality and heart failure hospitalisations even with a larger sample size.

Several smaller randomised trials investigated targeted left ventricular lead placement at the latest mechanical or electrically activated site of the left ventricle, and some reported a positive effect.^{10–18} The variable results across previous trials might reflect differences in study methods, including heterogeneity in trial design and outcome measures. Many studies used different methods to determine the latest site of activation and they relied on softer endpoints, such as echocardiographic variables or clinical risk scores. Less rigorous left ventricular lead implantation strategies in controls might have contributed to the observed discrepancies in outcomes. A medium-sized trial published in 2023 showed no effect of targeting a left ventricular pacing site where pacing resulted in acute reduction of electrical dyssynchrony.²⁶ Despite the close correspondence between mechanical contraction and electrical activation, it remains to be shown whether alternative left ventricular targeting strategies, such as targeting sites of late mechanical contraction, are beneficial to clinical endpoints.^{20,27}

	Intervention group (n=499)	Control group (n=501)	Total (N=1000)
Total complications	88	75	163
Major complications	71	54	125
Lead dislodgement	29	13	42
Left ventricular lead	16	5	21
Right atrial lead	8	4	12
Right ventricular lead	3	3	6
Not available	2	1	3
High threshold not resolved by reprogramming	4	3	7
Left ventricular lead	1	1	2
Left bundle branch lead	1	0	1
Right ventricular lead	2	2	4
Lead fracture or insulation defect	3	2	5
Lead perforation	3	0	3
Pneumothorax with drainage	2	2	4
Sensing failure not resolved by reprogramming	1	2	3
Phrenic nerve capture not resolved by reprogramming	6	8	14
Haematoma requiring reintervention and/or prolonged hospitalisation	2	0	2
Bleeding requiring reintervention	1	1	2
Local device infection	7	4	11
Systemic device infection*	2	6	8
Acute decompensation	1	0	1
Tamponade	4	3	7
Ventricular arrhythmias	3	2	5
Procedure-related death	1	0	1
Coronary sinus dissection, left ventricular lead not implanted	1	2	3
Other, major†	1	6	7
Minor complications	17	21	38
Pneumothorax without drainage	1	2	3
Haematoma without reintervention or prolonged hospitalisation	3	8	11
Superficial pocket infection treated with antibiotics	3	2	5
Coronary sinus dissection, left ventricular lead implanted	9	9	18
Other, minor‡	1	0	1

Data are n (%). *One patient was treated with antibiotics without device removal. †Pain at the device site with revision (n=3), bleeding (n=1), upper extremity deep venous thrombosis (n=1), right ventricular lead not connected to generator (n=1), damage to pre-existing right ventricular lead during implantation requiring reimplantation of a new lead (n=1). ‡Upper extremity oedema due to venous congestion without deep venous thrombosis (n=1).

Table 3: Procedure-related and device-related complications

CRT with biventricular pacing was introduced more than 30 years ago, and has undergone continuous refinement in patient selection, lead implantation techniques, and device programming. Although no singular technological advancement has independently shown a definitive benefit on clinical endpoints in adequately powered trials, the effect on cardiac function, symptoms, and mortality has improved remarkably over the years.^{28–30} This trial represents contemporary CRT implementation with quadripolar left ventricular leads and optimised programming of pacing parameters in patients receiving guideline-directed medical therapy for heart failure. We observed a substantial 14 percentage points absolute increase in mean LVEF after 6 months, which supports the durability and efficacy of contemporary biventricular pacing for heart failure. This improvement is larger than observed in most previous trials reporting remodelling after 6 months and can be used as a benchmark for contemporary CRT in heart failure, including conduction system pacing.^{2,3} Conduction system pacing might be difficult to achieve in patients with distal conduction system disease or interventricular septal scars, and attaining true conduction system capture can be challenging. Consequently, biventricular pacing will likely remain the treatment of choice for many patients with heart failure and wide QRS complex.

Strengths of the present trial include the nationwide collaboration among all five high-volume centres performing CRT implantations in Denmark, as reflected by the 98% success rate in achieving biventricular pacing. Additional strengths include its double-blind design, clinically relevant long-term follow-up, and the high number of observed events. With few exclusion criteria, this nationwide study population likely reflects an unselected cohort of patients eligible for CRT. Limitations include the selection of patients with a life expectancy of more than 2 years, the predominance of patients with LBBB or a paced QRS and few patients with other BBB QRS, missing data of lead placement on cardiac CT in a minority of patients, and anatomical limitations of the coronary sinus branches for left ventricular lead placement inherent to transvenous biventricular pacing. Furthermore, there was a high proportion of missing data for 6-min walk test and quality of life. In addition, algorithms of automatic optimisation of the atrioventricular delay to promote fusion of intrinsic conduction and left ventricular pacing were not used in the trial, and left ventricular pacing only was not allowed. Centre variability cannot be excluded and was not considered in the sample size calculation; however, the effect is likely modest as only five high-volume centres were involved in this trial. In support of no substantial effect of centre variability, the multivariable analysis for the primary endpoint, which yielded results similar to the modified intention-to-treat analysis, included centre as one of the variables. The choice of mapping under right ventricular pacing in patients with other BBB or an indication for ventricular pacing introduces methodological

heterogeneity; however, this was unlikely to differ between group. Finally, we used the inter-lead delay between the two ventricular pacing leads in assessing the delay to left ventricular activation, which is different from the QLV measured from onset QRS to the activation on the left ventricular lead often used in the observational literature reporting better outcomes in patients with longer QLV.

In patients with heart failure and prolonged QRS, targeted placement of the left ventricular pacing lead at the site of latest electrical activation did not reduce the risk of the composite endpoint of death or unplanned hospitalisation for heart failure compared with conventional placement of the lead in a posterolateral, non-apical position.

Contributors

Conceptualisation: JCN, JHS, JBJ, JHa, LK, CGe, JK, AMS, and MBK. Data curation: JHj, RGS, MHJPF, EAF, and NR. Verification of raw data: JHj, RGS, and JCN. Formal analysis: RGS was the trial statistician primarily responsible for the statistical analysis, contributions from JHj, MHJPF, JCN, MBK, EAF, and NR. Funding acquisition: JCN. Investigation: JCN, JHS, JBJ, JML, CEL, SHP, HE, BLN, NR, LK, JK, KØ, AMS, MS, PS, MKP, KFK, TFH, MJM, MVin, HKJ, JHa, TZ, MVie, BP, CGe, and MBK all contributed to patient inclusion or/and treatment and follow-up of patients in the trial. Methodology: JCN, JHS, JBJ, JHa, MBK, LK, NR, and CGr. Project administration: JCN and MBK coordinated the multicentre trial and monitored and reported trial progress, timelines, budgets, and compliance with ethical standards as primary investigator and coprimary investigator. JCN, JHS, JBJ, JHa, MBK, CEL, JML, and NR have been responsible for logistics and organising conduct of the trial in their institutions and supervision of trial personnel. BLN, KFK, NR, TZ, and AMS planned and organised cardiac CT data acquisition and analysis. SHP, NR, KØ, PS, and TFH were responsible for organising acquisition and analysis of echocardiographic data. CGr was responsible for planning, organising, and coordinating the multicentre acquisition of digital electrocardiograms throughout the study period. HE, LK, MKP, MS, and PS were responsible for coordinating and supervising heart failure management during the trial conduct. Resources: JHj was responsible for the digital case-record form including the randomisation module used in the trial and for the data storage solution throughout trial conduct. Visualisation: MHJPF, RGS, and EAF created figures. Original manuscript drafting: JCN with contribution from MBK and MHJPF. All authors had access to the data, contributed to the review and revision of the manuscript, approved the final version, and had final responsibility for the decision to submit for publication.

Declaration of interests

JCN is Executive Editor for *Europace*. JHS received travel grants from Medtronic and Abbott; is an Advisory Board Member for Medtronic and Vital Beats; and holds Vital Beats stock options. JBJ received personal honoraria from Medtronic and Boston Scientific and is Chairman of the Danish Pacemaker and ICD Registry. LK received speaker's honorarium from AstraZeneca, Bayer, Boehringer, Novartis, and Novo Nordisk. MS received lecture fees from Novo Nordisk, AstraZeneca, Boehringer, and Bayer. KFK received institutional research grants from AP Møller og hustru Chastine McKinney Møllers Fond, the Novo Nordisk Foundation, Novo Nordisk, Sygeforsikringen Danmark, Canon Medical Systems, and GE Healthcare. MHJPF is member of the European Heart Rhythm Association Scientific Documents Committee and Medtronic Advisory Board; and received consulting fees from Medtronic. EAF received grants from Rigshospitalets Forskningspulje, Fonden af 17–12–1981, Eva og Henry Frønkels Mindefond, and Arvid Nilssons Fond for DANISH-CRT substudies. MVin received a presentation fee from Medtronic; received an institutional grant from Abbott for attending EHRA 2026; and is a member of Medtronic Advisory Board for Conduction System Pacing. HKJ received grants from the Novo Nordisk Foundation (NNF18OC0031258 and NNF20OC0065151). MVie received personal honoraria for lecture from Medtronic. JHj is part of the Data

Safety Monitoring Board for this trial. MBK received speakers bureaus fees from Biosense Webster and Abbott. All other authors report no competing interests.

Data sharing

All study-related documents will be made available on request to the corresponding author (jenniels@rm.dk). Individual data collected for the study will be made available for collaborative pooled analyses provided relevant contracts and data sharing agreements are made. Only anonymised data will be shared.

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